

W5 L2

Feb 22, 2024



CFG

G:

$$A \rightarrow 0A1$$

$$A \rightarrow B$$

$$B \rightarrow \#$$

$L(G)$

$$= \{0^n \# 1^n \mid n \geq 0\}$$



$$A \Rightarrow 0A1 \Rightarrow \overline{0} \overline{B} \overline{1} \Rightarrow \underline{\underline{0 \# 1}}$$

$$A \Rightarrow^* 0 \# 1$$

$$\Sigma = \{0, 1\}$$

$$R_1 = (0+1)^k$$

$$R_2 = (1+0)^k$$

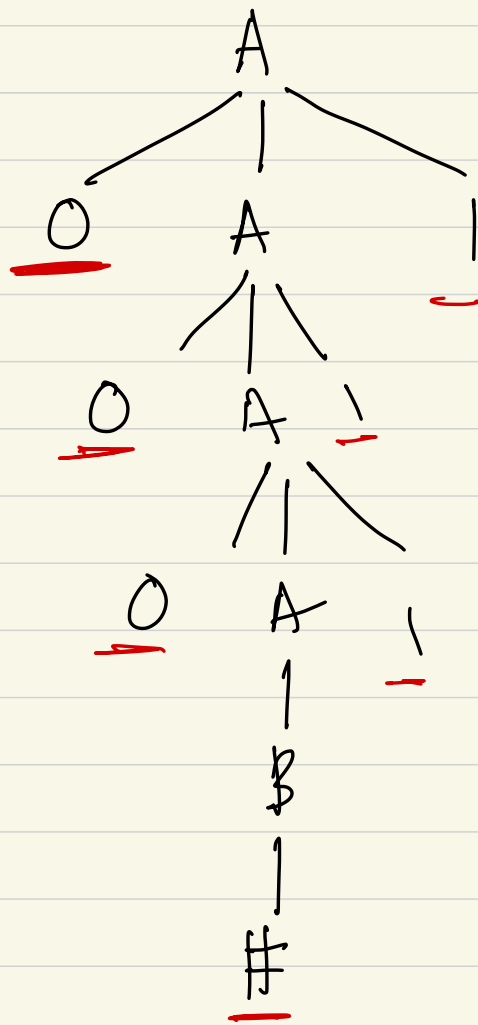
$$1+2$$

$$2+1$$

$$3 \times 1$$

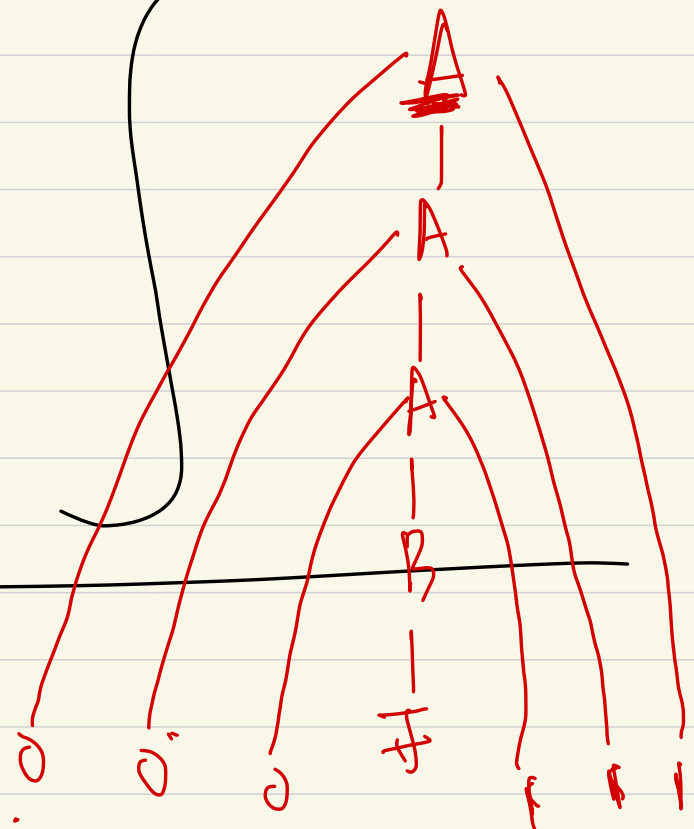
$A \Rightarrow 0A1 \Rightarrow 00A11 \Rightarrow 000A111$

$\Rightarrow 000B111 \Rightarrow 000\#111$



Parse
Tree

000#111



$$G = (V, \underline{\Sigma}, R, \underline{\langle \text{expr} \rangle}) \quad \Sigma = \{a, +, *, (,)\}$$

$$V = \{ \langle \text{expr} \rangle, \langle \text{term} \rangle, \langle \text{factor} \rangle \}$$

$$R = \{$$

$$\langle \text{expr} \rangle \rightarrow \langle \text{expr} \rangle + \langle \text{term} \rangle$$

$$\langle \text{expr} \rangle \rightarrow \underline{\langle \text{term} \rangle}$$

$$\langle \text{term} \rangle \rightarrow \langle \text{term} \rangle * \langle \text{factor} \rangle$$

$$\langle \text{term} \rangle \rightarrow \underline{\langle \text{factor} \rangle}$$

$$\langle \text{factor} \rangle \rightarrow (\langle \text{expr} \rangle)$$

$$\langle \text{factor} \rangle \rightarrow a$$

}

$$L(G) = \{ "a+a", "a*a", "(a)", "(a+a*a)", \\ "a+(a+a)", "a*(a+(a+a))", \\ \dots \}$$

$$\left[\underbrace{a}_{\text{expr}} + \underbrace{a*a}_{\text{term}} \right]$$

$$\langle \text{expr} \rangle \Rightarrow \underline{\langle \text{expr} \rangle} + \underline{\langle \text{term} \rangle}$$

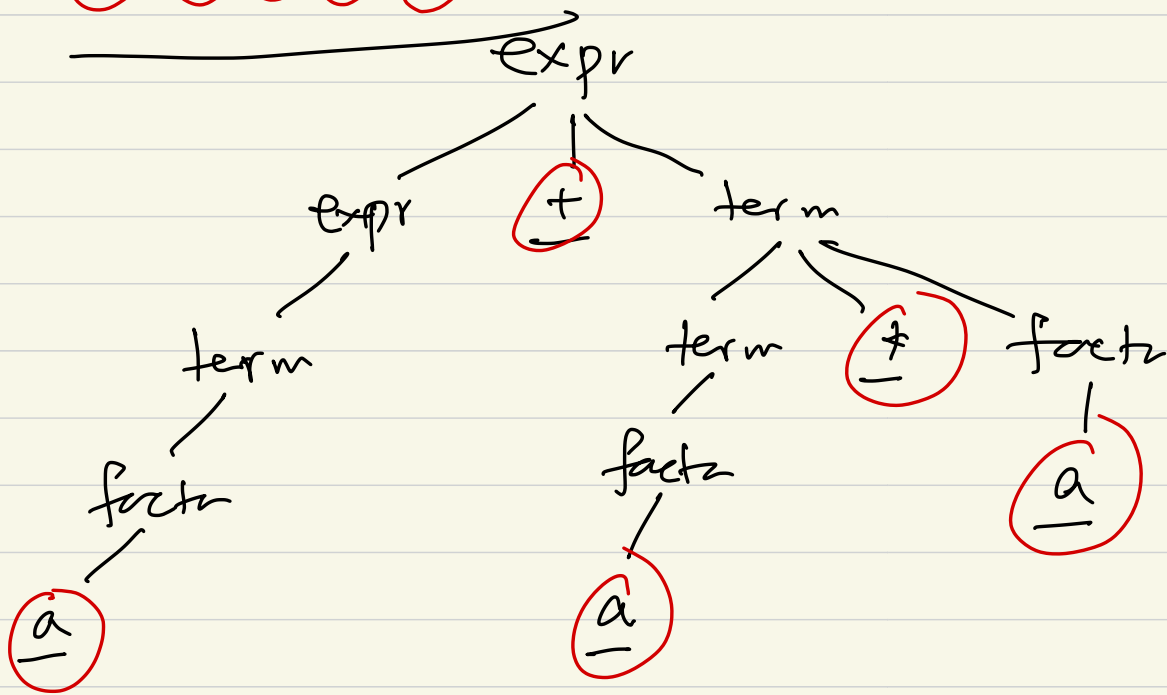
$$\Rightarrow \underline{\langle \text{term} \rangle} + \langle \text{term} \rangle \Rightarrow \underline{\langle \text{factor} \rangle} + \langle \text{term} \rangle$$

$$\Rightarrow \underline{a} + \underline{\langle \text{term} \rangle} \Rightarrow a + \underline{\langle \text{term} \rangle} * \langle \text{factor} \rangle$$

$$\Rightarrow a + \langle \text{factor} \rangle * \langle \text{factor} \rangle \Rightarrow a + a * \langle \text{factor} \rangle$$

$$\Rightarrow \underline{a + a * a}$$

a + a * a



$\langle \overset{E}{\text{expr}} \rangle \rightarrow \langle \text{expr} \rangle + \langle \text{expr} \rangle \mid$

$\langle \text{expr} \rangle * \langle \text{expr} \rangle \mid$

$(\langle \text{expr} \rangle) \mid$

ϵ

a + a * a

$E \Rightarrow E + E \Rightarrow E + E \Rightarrow a + E$

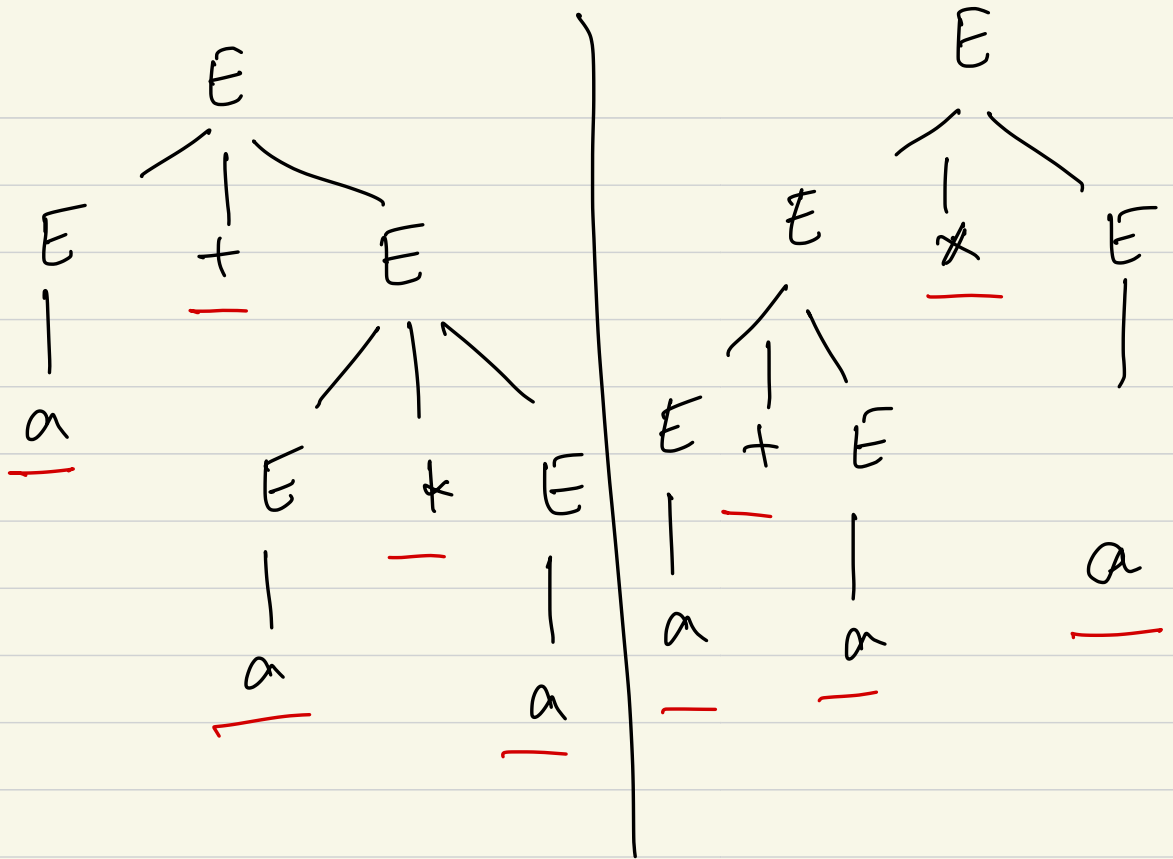
$\Rightarrow a + E + E \Rightarrow a + a * E$

$\Rightarrow \underline{a + a * a}$

$E \Rightarrow E * E \Rightarrow E + E * E \Rightarrow a + E * E$

$\Rightarrow a + a * E \Rightarrow a + a * a$

①



```

if (condition1) {
    if (condition2) {
        _____
    }
    else {
        _____
    }
}

```

```

if (condition1) {
    if (condition2) {
        _____
        .
        .
        .
    }
    else {
        .
        .
        .
    }
}

```

```
int    x, y;  
if (x == y) {  
    print ("Hello");  
    if (x > 0) {  
        print (" world!");  
    }  
    else {  
        print (" and bye");  
    }  
}
```